

# FaultBridge

A code-switched Nigerian telco support agent that grounds answers in verified operational data, resolves eligible faults, and turns repeated unresolved complaints into unconfirmed incidents for NOC review.

### EVALUATION QUESTION

Can a voice agent understand Hausa-English, Igbo-English, Nigerian Pidgin-English, and Yoruba-English while choosing the correct tools, protecting caller data, and avoiding invented network facts?

400 natural clips

4 ASR systems

68.63 minutes

351 source groups

55.0%

Sahara equal-language WER, the strongest aggregate on the frozen natural-speech panel.

## Section 1: Natural code-switched ASR

### DATA

100 hash-frozen AfriSwitch clips per language pair (400 total, 68.63 minutes), balanced across code-mixing index, switch count, duration, and source groups. All audio resampled to 16 kHz mono PCM16. No transcript normalisation beyond Unicode NFKC and whitespace collapsing. Every model received identical audio. Failed and empty outputs stayed in the denominator. Confidence intervals use 2,000 deterministic source-clustered bootstrap samples.

### QUANTITATIVE RESULTS: WER PER LANGUAGE

| Language pair      | Sahara | SBPN  | OmniASR | Faster-Whisper |
|--------------------|--------|-------|---------|----------------|
| Hausa-English      | 38.9%  | 45.1% | 68.0%   | 98.3%          |
| Igbo-English       | 62.1%  | 65.7% | 79.9%   | 85.0%          |
| Pidgin-English     | 35.6%  | 39.3% | 55.7%   | 51.5%          |
| Yoruba-English     | 83.6%  | 82.1% | 81.4%   | 91.2%          |
| Equal-language WER | 55.0%  | 58.1% | 71.2%   | 81.5%          |

WER = word error rate (lower is better). Bold values mark the best-performing model per row. Empty-output rates: Sahara 10.5%, SBPN 8.8%, OmniASR 4.5%, Faster-Whisper 4.0%. Eight difficult clips were empty for every model.

### QUALITATIVE FINDINGS PER MODEL AND LANGUAGE

| Model          | Strengths                                                                                                                                                                                     | Weaknesses                                                                                                                                                                                                                                 |
|----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sahara         | Lowest WER on Hausa (38.9%), Igbo (62.1%), and Pidgin (35.6%). Highest code-switch recall on Pidgin (40.2%). Conservative error profile: deletes uncertain content rather than hallucinating. | Highest empty-output rate (10.5%). Yoruba deletion rate reaches 47%, losing nearly half the local-language content. Yoruba WER (83.6%) is not statistically better than SBPN or OmniASR.                                                   |
| SBPN           | Fastest latency (p50: 0.54s). Near-zero Pidgin language-role gap, suggesting specific tuning for Nigerian Pidgin phonemics. Balanced error profile across language roles.                     | Consistently 3-4 pp behind Sahara on Hausa, Igbo, and Pidgin. Higher deletion on Yoruba (28%). Cannot match Sahara's switch-context recall on any pair except Pidgin.                                                                      |
| OmniASR        | Fewest empty outputs (4.5%). Marginally best Yoruba WER (81.4%), though statistically inconclusive. CTC architecture provides stable output.                                                  | Language-agnostic CTC ignores code-switch conditioning. Severe English-span errors on Hausa (85.1%). Minimal switch preservation across all pairs (2.5-22.7%).                                                                             |
| Faster-Whisper | Lowest empty-output rate (4.0%). Performs reasonably on Pidgin (51.5%) where creole structure helps.                                                                                          | Catastrophic Hausa failure (98.3% WER). VAD filter tradeoff: disabling it produces text but with hallucinations and boilerplate. High substitution rates suggest the decoder invents content under uncertainty rather than staying silent. |

**Key pattern.** Pidgin is easiest for all models (creole structure, shared English vocabulary). Yoruba is hardest (tonal phonology, high deletion). Model ordering is stable across pairs except Yoruba, where all models plateau at 81-91% WER.

200/200

Sahara TTS outputs generated

648

real model + PostgreSQL agent runs

0%

critical safety failure rate

SAHARA CODE-SWITCHED TTS

100 genuine code-switched AfriSwitch texts (25 per language) synthesized with female and male Sahara voices. All outputs were valid 22.05 kHz mono PCM16 WAVs, totaling 1,672 seconds. Three independent ASR families judged every output.

| Independent judge             | WER   | Extra words | Transcript loss | Segment loss |
|-------------------------------|-------|-------------|-----------------|--------------|
| SBPN Multilingual Base        | 39.7% | 2.2%        | 7.6%            | 27.7%        |
| Meta OmniASR CTC 300M v2      | 51.3% | 2.1%        | 9.4%            | 36.8%        |
| Faster-Whisper large-v3-turbo | 58.6% | 3.3%        | 12.7%           | 39.6%        |

Automatic audit signals, not confirmed TTS defects. Pidgin had strongest judge consensus (2.4 pp spread, SBPN WER 21.9%). Yoruba had weakest (all judges above 67%, segment loss 50-60%). No meaningful gender difference (female 51.3% vs. male 51.3% overall). A frozen 40-file bilingual listening audit is prepared before any MOS claim.

QUANTITATIVE RESULTS: DOWNSTREAM TASK PERFORMANCE PER ASR MODEL

48 scenarios (12 per language) covering fault lookup, diagnosis, escalation, and safety paths. Together-hosted Llama 3.3 70B with deterministic policy, real typed tools, and isolated PostgreSQL. Gold panel: 288/288 passed. Latency: 1.20s p50 / 2.25s p95.

| 24-utterance named-ASR panel | Strict task success | Assertion accuracy | Capability retained |
|------------------------------|---------------------|--------------------|---------------------|
| Gold transcript              | 100.0%              | 100.0%             | 100.0%              |
| Sahara ASR                   | 100.0%              | 100.0%             | 100.0%              |
| SBPN                         | 95.8%               | 99.2%              | 95.8%               |
| OmniASR                      | 87.5%               | 98.4%              | 87.5%               |
| Faster-Whisper               | 79.2%               | 97.6%              | 79.2%               |

FAILURE ANALYSIS

All 27 strict failures came from nine stable scenario/provider cells where ASR lost a symptom keyword or language cue. Root causes: lost complaint classification (e.g. "balance na finish" truncated), missed code-switch causing wrong tier selection, and diacritic loss causing database lookup failure. All failures defaulted to escalation, never to unauthorised action.

PRIVACY GATE

100 synthetic cases with 100 independently labelled spans (25 phone, 25 email, 25 account/SIM, 25 numeric): 100% exact typed-span precision, recall, and F1. Zero post-redaction leakages. Out of scope: personal names, street addresses, word-spelled numbers.

EVALUATION METRICS AND RATIONALE

- **WER** = substitutions + deletions + insertions / reference words. Standard ASR metric. May exceed 100%.
  - **Hallucination / transcript loss** = insertion and deletion rates as automatic TTS fidelity indicators.
  - **Segment loss** = share of tagged language spans with no correctly aligned word. Captures whole-span failure.
- **Strict task success** = every analysis, tool-trace, response, state, and privacy assertion must pass. Measures whether ASR error changes the agent's action.
  - **Capability retention** = ASR-transcript task success / gold-transcript success. Isolates ASR-induced propagation loss.
  - **Exact typed-span F1** = precision and recall on character-exact PII boundaries per type.

MEASURED ROUTING DECISION

**Use Sahara for all four language pairs in the submission.** No alternative currently combines a statistically supported paired WER advantage, fewer than 1% provider failures, comparable deployed p95 latency, and zero critical downstream failures. The evidence-gated router is implemented, but multi-ASR routing stays off until those conditions pass.

QUALITATIVE SUMMARY PER LANGUAGE PAIR

| Language       | Difficulty | Best model     | Key finding                                                                                                                                                                                                                            |
|----------------|------------|----------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Hausa-English  | Medium     | Sahara (38.9%) | Widest model spread (59.4 pp between best and worst). Faster-Whisper collapses entirely (98.3%). English spans inside Hausa are harder to recognise than Hausa itself. Sahara's switch-context recall (30.5%) is highest among models. |
| Igbo-English   | Hard       | Sahara (62.1%) | All models show elevated deletion of Igbo content. Switch preservation drops to 18.2% (Sahara), suggesting Igbo code-switches are acoustically subtler. SBPN achieves balanced language-role errors.                                   |
| Pidgin-English | Easiest    | Sahara (35.6%) | Creole structure and shared English vocabulary help all models. Highest switch recall (40.2% Sahara, 39.9% SBPN). Strongest TTS judge consensus (2.4 pp spread). SBPN achieves near-zero language-role gap.                            |
| Yoruba-English | Hardest    | Inconclusive   | All models plateau at 81-91% WER. Tonal phonology causes systematic deletion (47% for Sahara). Switch preservation below 6% for all. TTS segment loss reaches 50-60%. Confidence intervals overlap, so no model is clearly best.       |

RESPONSIBLE DEPLOYMENT BOUNDARY

**Consent and privacy.** Declined consent bypasses speech and model providers. Accepted transcripts are redacted before the LLM, database, tools, logs, or UI. Caller IDs are HMAC-pseudonymised. Raw audio is handled in memory only.

**Grounding.** Verified incidents and account records enter through authenticated internal APIs. Missing values stay unknown.

**Agent authority.** The LLM extracts a constrained complaint schema. Deterministic policy authorises typed tools, and every tool input/output is audited in PostgreSQL.

**Crowd discovery.** Three distinct callers may create an *unconfirmed* candidate incident. Only an operator can verify it. Crowd signals are never presented as confirmed outages.

REPRODUCIBLE EVIDENCE

- Frozen manifests bind every sample and generated WAV to SHA-256.
- Raw JSONL is append-only and retains failures, provider IDs, model versions, code commit, lock hash, and latency.
- make benchmark-verify-asr-evidence, -tts-evidence, and -agent-evidence regenerate public aggregates byte for byte.
- Private ASR evidence: Tiamz/faultbridge-asr-benchmark-evidence, revision ef320d7f...fe00.

LIMITATIONS

The natural ASR panel is difficult research speech, not a population prevalence sample. The named-ASR panel uses one Sahara female synthetic voice and may favour Sahara recognition. Automatic TTS judges cannot replace bilingual listening. The privacy claim excludes names, street addresses, and word-spelled numbers. API, CPU, and GPU timing environments are not directly comparable. FaultBridge still requires authoritative NOC/CRM integrations, a telco security review, matched live-stream latency, and human call trials before production.

**Sources and artifacts.** AfriSwitch dataset card: [huggingface.co/datasets/intronhealth/AfriSwitch](https://huggingface.co/datasets/intronhealth/AfriSwitch) · Intron speech API: [docs.voice.intron.io](https://docs.voice.intron.io) · RAI speech evaluation: [isca-archive.org/interpeech\\_2025/rai25\\_interpeech.pdf](https://archive.org/interpeech_2025/rai25_interpeech.pdf) · Full methods: [docs/ASR\\_BENCHMARK\\_REPORT.md](#), [docs/TTS\\_BENCHMARK\\_REPORT.md](#), [docs/AGENT\\_BENCHMARK\\_REPORT.md](#), [docs/PRIVACY\\_BENCHMARK\\_REPORT.md](#).